

Modelling Frameworks

Today's agenda:

How ecologists choose models

Where variability comes from

Software installation and setup

Why not just run a correlation?

How would you analyze this?

- Reef fish in the Caribbean. You have population counts through time, and underwater video of predator visits.
- Question: does living at high density make a fish more likely to be eaten?
- How would you test this based on what you already know?

Why not just run a correlation?

Challenges to deal with:

- Correlation assumes bivariate normal data.
- Regression assumes a linear relationship and normal errors.
- Mortality counts are neither (also there were lots of zeros)

Why not just run a correlation?

What Sandin & Pacala (2005) did instead:

Wrote the model down directly:

- absolute mortality = $r_0 + r_1n$
- per capita mortality = $(r_0 + r_1n) / n$
- with binomial and Poisson variation instead of normal

Why not just run a correlation?

What Sandin & Pacala (2005) did instead:

This yields predator visits per hour per additional prey individual.

This number has *ecological units* with real meaning

(not just a p-value or a slope on transformed data.)

This course will help you add skills like this to your toolbox

(but also correlations and linear regressions)

Learning to model

"I want to learn how to model"

Too broad to act on.

Compare: "I want to learn genetics."

Population structure? Gene expression? Phylogenetics?

Which question are you actually asking?

Learning to model

This course covers one kind

Statistical modelling, leaning mechanistic

Lots of other modelling frameworks

Sometimes frameworks can be combined

What you should already know

What you should already know:

Statistics: basic familiarity (e.g., one undergrad semester)

Computers: beyond point-and-click

Math: rusty calculus is fine

Ecology: basics, e.g., growth, functional responses

Which is rustiest?

Statistics? Computers? Math? Ecology?

The book lists remedial reading for each

Iteration and repetition

We will often cover things multiple times

To see why details matter, you need the big picture

To grasp the big picture, you need some details

This also helps with long term memory

What are some of the model choices mentioned?

Scope and approach

What is the model for?

abstract ↔ concrete

strategic ↔ tactical

general ↔ specific

theoretical ↔ applied

qualitative ↔ quantitative

descriptive ↔ predictive

mathematical ↔ statistical

mechanistic ↔ phenomenological

pattern ↔ process

Note: Generality has a price

Theoretical models: general, oversimplified

Written in precise numbers of individuals, but used for qualitative predictions

Applied models: messier, more specific, better at actual forecasts, harder data requirements

Technical details

How is it built and analyzed?

analytical $\leftrightarrow$ computational

dynamic $\leftrightarrow$ static

continuous $\leftrightarrow$ discrete

population-based $\leftrightarrow$ individual-based

deterministic $\leftrightarrow$ stochastic

Dynamic vs. Static Models

Dynamic Models

Most mathematical models

Today's state feeds tomorrow's

Static Models

Most statistical models

The relationship is fixed

Combining them is hard (Chapter 11)

Sophistication

Simple ↔ Complex

How long is the description?

Crude ↔ Sophisticated

How much conceptual depth is hidden?

Note on Model Classification

A model can be simple and sophisticated at once

Note: Simple is not crude

Virtual Coin Flips

Simple, crude

rbinom()

Simple, sophisticated

Random Walk

Simpler still, more sophisticated

predicts any birth/death rate
without simulating

Sophistication buys power, costs transparency

Why are there so many different options?

Why are there so many different options?

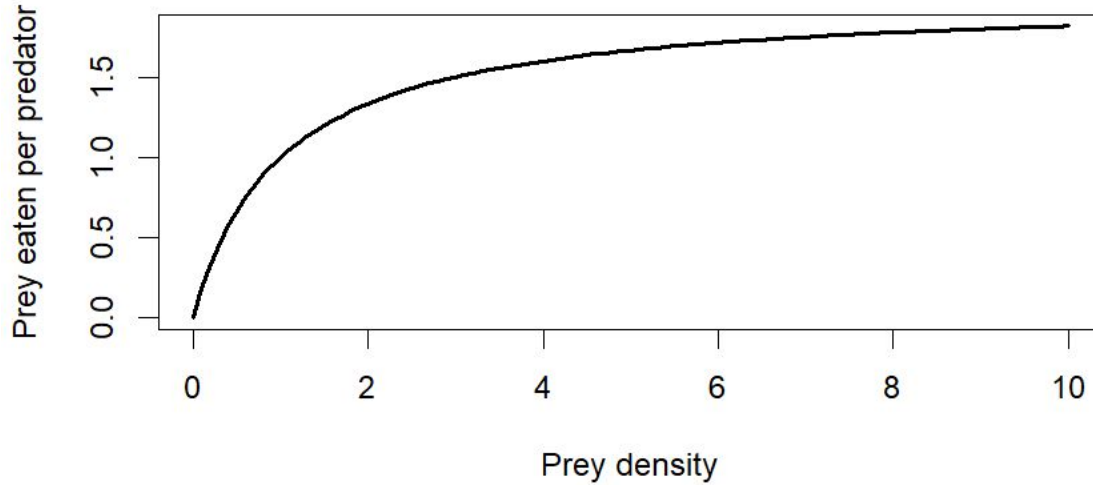
Trade-offs: models differ in when they're useful

E.g., mechanistic models can be linked to specific mechanisms,
but may require more data!

You'll have to consider these trade-offs when selecting methods

Questions?

A shape you've seen before



Describe it in words.

Where else have you seen this shape?

Two ways to end up here

A. I needed a curve that starts at zero, rises, then flattens. This one does.

B. A predator can only handle so many prey per hour.

Mechanistic vs Phenomenological. Which is which?

Same curve, different claim

$$f(x) = ax / (b + x)$$

This version has a name: Holling type II functional response

Here, a and b are attack rate and handling time.

You can only interpret a and b that way if that's why you chose the curve.

Why does it matter?

Chose it for its shape?

You can *describe* your data.

Chose it for the mechanism?

You can talk about the *process, extrapolate further*, and be wrong in a specific way.

Two parts to every model



Deterministic

The expected pattern

The structured, predictable core relationships modeled by mathematical formulas.



Stochastic

How observations scatter

The random, unpredictable variation or noise superimposed on top of the system's deterministic trend.

We've done the first part. **Where does the scatter come from?**

Three Kinds of Randomness



Measurement Error

Observation Noise



Demographic

Individual Heterogeneity



Environmental

External Heterogeneity

Measurement error

Observations are (usually) imperfect

Measurements have intrinsic limits (e.g., nearest m or mm)

Almost always present

Usually modelled as normal variation around a mean

Demographic stochasticity

Different outcomes, by chance alone

One coin, 20 flips: 9 heads. Or 10. Or 11.

50% survival of 9 individuals: 4 or 5 survive, never 4.5

Matters most in small populations

20 identical tadpoles, one predator, a different number eaten each time

Environmental stochasticity

Variability from outside the system

Climate, season, topography

Only the unpredictable part (predictable seasonal change is deterministic)

Process vs. measurement

Demographic + environmental stochasticity = process variability

Process variability changes what happens next in a system

Measurement error doesn't

Process vs. Measurement: Why It Matters

The scenario: A plot has 12 seedlings.

Measurement Error

What happens:

You miscount and record 9.

The consequence:

Recount next week: find 12.

Process Error

What happens:

A deer eats 3.

The consequence:

Recount next week: 9, and 9 forever.

Error types: your turn

Sort these into measurement / demographic / environmental :

1. Two observers, same transect, different counts
2. A hurricane kills a seedling cohort
3. 50 tagged fish: 12 survive, then 15
4. A thermometer reads 0.4 °C higher than true
5. Seedlings differ with small-scale light

Your own system

**Think of one question you'd like to ask in your system.
What type of model could you use to address it?**

theoretical / applied

mechanistic / phenomenological

deterministic / stochastic

Questions?

Our goal for the remaining time is setting up software

We'll be installing:

R

RStudio

Git (optional for undergrads)

Github (optional for undergrads)

R

```
FOREST+bmaitner@USFAAD-228TFW3 MINGW64 ~/Desktop/current_projects/BIEN_4.2_update_ms (main)
$ R
```

```
R version 4.4.2 (2024-10-31 ucrt) -- "Pile of Leaves"
Copyright (C) 2024 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64
```

```
R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.
```

```
  Natural language support but running in an English locale
```

```
R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.
```

```
Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.
```

```
> library(BIEN)
Loading required package: RPostgreSQL
Loading required package: DBI
Type vignette("BIEN") or vignette("BIEN_tutorial") to get started
>
```

RStudio

The screenshot displays the RStudio IDE interface. The top menu bar includes File, Edit, Code, View, Plots, Session, Build, Debug, Profile, Tools, and Help. Below the menu is a toolbar with icons for saving, running, and other functions. The main editor window shows a script titled '1_BIEN_stats.R' with the following R code:

```
1 # Code to generate BIEN annotation statistics
2
3 library(BIEN)
4 library(tidyverse)
5
6 # Total number of records in view_full_occurrence_individual
7
8 total <- BIEN:::.BIEN_sql(query = "SELECT count(*)
9                                FROM view_full_occurrence_individual ;")
10
11 #284466171 # Total number of records in vfoi
12
13
14 # Total of different observation types
15
16 occurrence_type_breakdown <-
17 BIEN:::.BIEN_sql(query = "SELECT observation_type, COUNT(*) AS total
18                          FROM view_full_occurrence_individual
19                          GROUP BY observation tvpe")
20
```

The console at the bottom shows the R version (4.4.2) and the current project path (C:/Users/bmaitner/Desktop/current_projects/BIEN_4.2_update_ms/). The file explorer on the right shows the project structure, including files like .gitignore, .Rhistory, 1_BIEN_stats.R, 2_BIEN_metadata.R, 3_UNESCO_use_case.R, 4_BIEN_complete_citations.R, BIEN_4.2_update_ms.Rproj, citations, figures, LICENSE, and output.

Environment History Connections Git Tutorial

R Global Environment

Data

total 1 obs. of 1 variable

Files Plots Packages Help Viewer Presentation

Folder File Delete Rename

C:\Users\bmaitner\Desktop\current_projects\BIEN_4.2_update_ms

Name	Size	Modified
..		
.gitignore	69 B	Feb 20, 2025, 12:07 PM
.Rhistory	0 B	May 2, 2025, 3:17 PM
1_BIEN_stats.R	22.7 KB	Feb 20, 2025, 11:48 AM
2_BIEN_metadata.R	208 B	Jan 24, 2025, 9:57 AM
3_UNESCO_use_case.R	2.8 KB	Feb 20, 2025, 11:26 AM
4_BIEN_complete_citations.R	1.8 KB	Jan 24, 2025, 9:57 AM
BIEN_4.2_update_ms.Rproj	267 B	Aug 26, 2025, 10:04 AM
citations		
figures		
LICENSE	1.1 KB	Jan 24, 2025, 9:57 AM
output		

Console Terminal Background Jobs

R 4.4.2 · C:/Users/bmaitner/Desktop/current_projects/BIEN_4.2_update_ms/

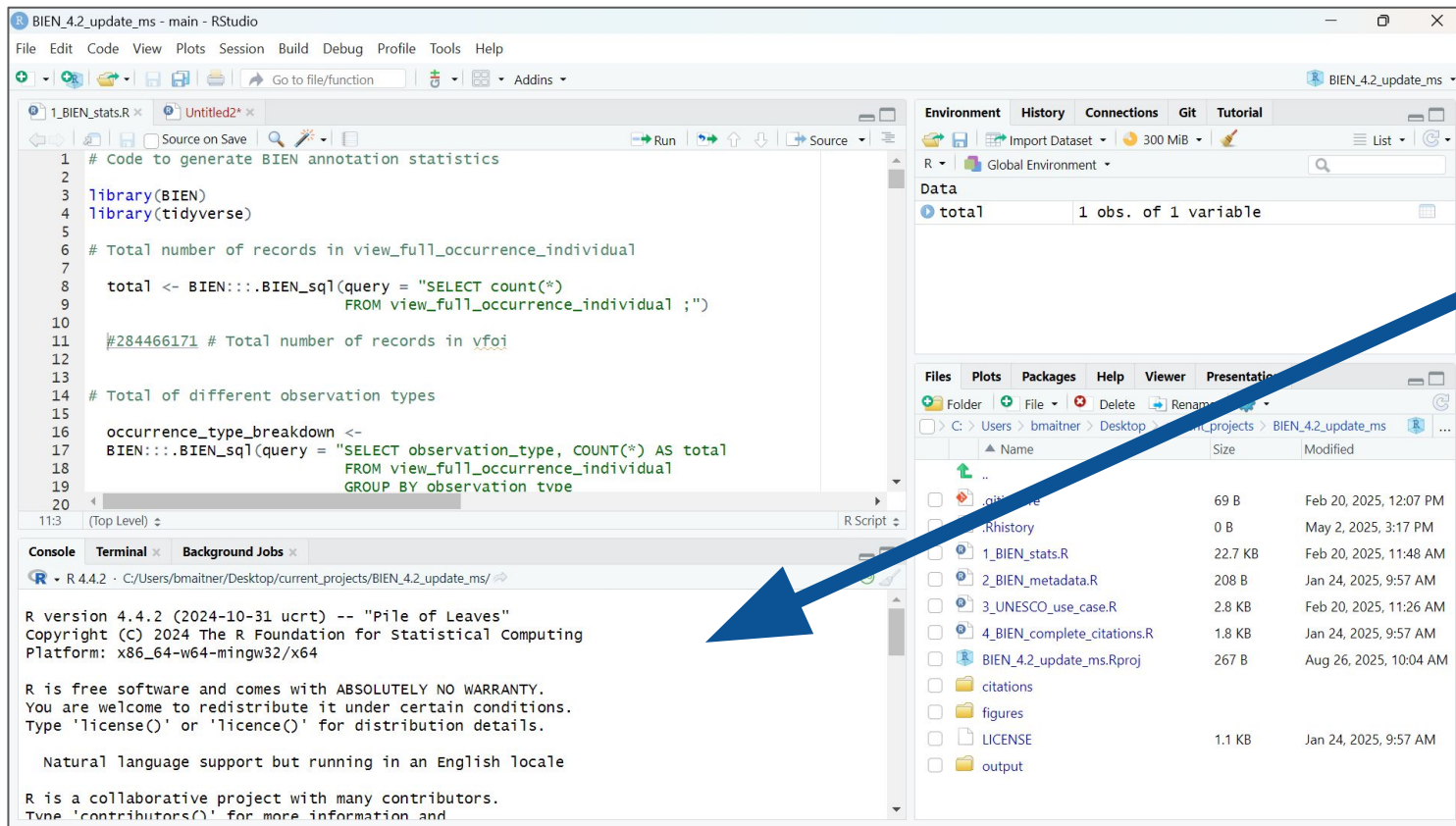
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RStudio



The screenshot displays the RStudio IDE interface. The main editor window shows a script titled "1_BIEN_stats.R" with R code for generating BIEN annotation statistics. The code includes library calls for BIEN and tidyverse, a SQL query to count records in the view_full_occurrence_individual table, and a breakdown of observation types. The console at the bottom shows the R version (4.4.2) and the R Foundation for Statistical Computing logo. The file explorer on the right shows the project structure, including files like 1_BIEN_stats.R, 2_BIEN_metadata.R, 3_UNESCO_use_case.R, 4_BIEN_complete_citations.R, BIEN_4.2_update_ms.Rproj, citations, figures, LICENSE, and output.

```
# Code to generate BIEN annotation statistics
library(BIEN)
library(tidyverse)

# Total number of records in view_full_occurrence_individual
total <- BIEN::BIEN_sql(query = "SELECT count(*)
                                FROM view_full_occurrence_individual ;")
#284466171 # Total number of records in vfoi

# Total of different observation types
occurrence_type_breakdown <-
  BIEN::BIEN_sql(query = "SELECT observation_type, COUNT(*) AS total
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R

RStudio

Stuff to make R easier to use

The screenshot shows the RStudio interface with the following components:

- Source Editor:** Contains R code for generating BIEN annotation statistics. Red arrows point to the code area and the `library(tidyverse)` line.
- Environment Pane:** Shows the current environment with a variable `total` of type `double` and value `1`. A red arrow points to this pane.
- Files Pane:** Shows the project file structure. A red arrow points to the `BIEN_4.2_update_ms.Rproj` file.
- Console:** Displays the R version (4.4.2) and the R Foundation's copyright notice.

Red arrows point to the following elements:

- The source editor (code area).
- The `library(tidyverse)` line in the source editor.
- The Environment pane.
- The Files pane.

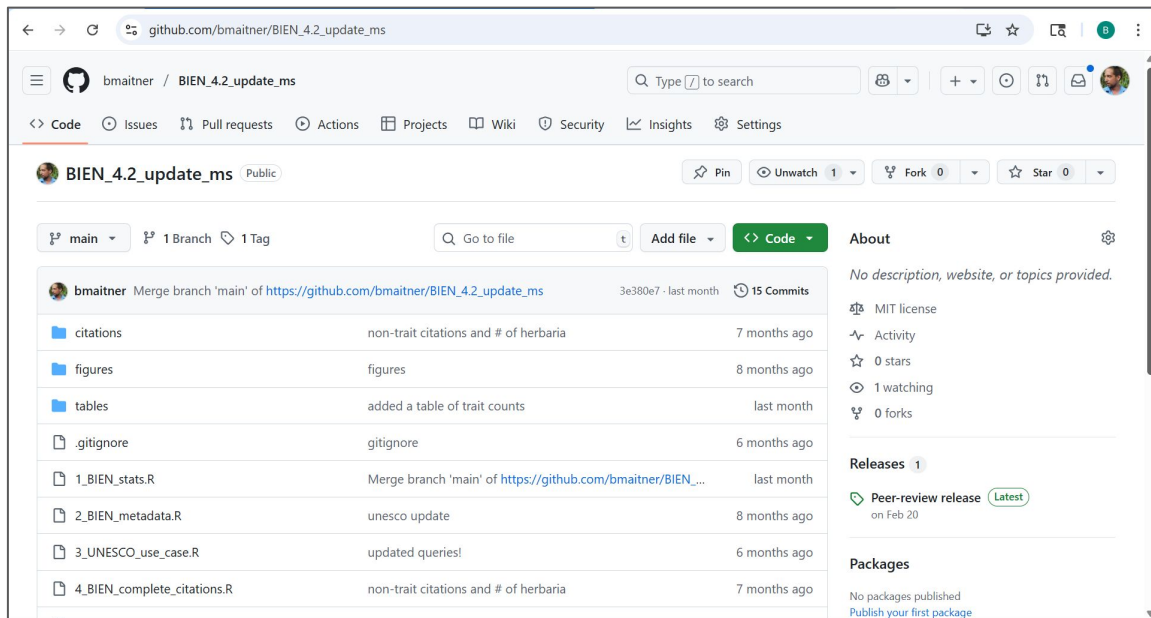
A large blue arrow points from the right side of the image towards the RStudio interface, with a large blue letter **R** next to it.

Git

- Program that tracks changes in files
- Removes need for multiple versions of file (you can always go back)
- Tracks what is added or removed in each version
- Allows multiple users to work on files and combine their changes

Github

- A repository for Git
- Stores your files
- Also can be used for editing files
- You can even set it up to run analyses for you!



Why use Github?

- Secure backup of code (in case your computer dies...)
- Helps when you use multiple computers
- Collaboration
- Transparency and reproducibility (if you make it public)
- Letting other people use your code (forking/copying)
- Increasingly expected of biologists
- Can be archived, cited (e.g., via Zenodo)
- Grad students have to, undergrads get extra credit

Installing the needed software

- R (free at <https://www.r-project.org/>)
- RStudio (free at <https://posit.co/download/rstudio-desktop/>)
- Git (free at <https://git-scm.com/downloads>)
- Github account (Sign up for free at: <https://github.com/>)

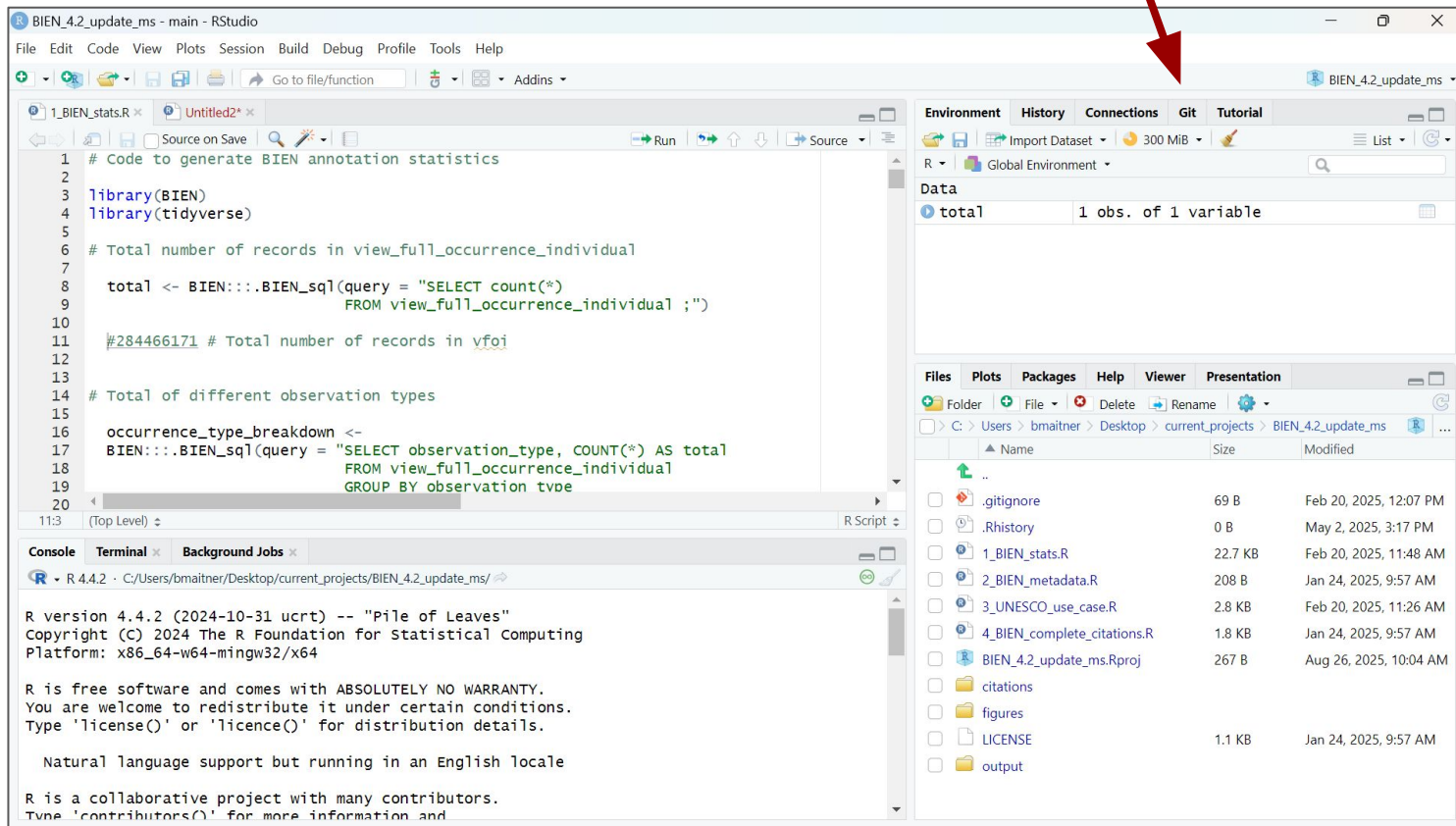
Tell Git who you are

```
install.packages(c("usethis", "gitcreds"))
```

```
usethis::use_git_config(  
  user.name = "Your Name",  
  user.email = "your GitHub email"  
)
```

Did it work?

Should have a “Git” tab in RStudio



The screenshot shows the RStudio interface with the following components:

- Source Editor:** Contains R code for generating BIEN annotation statistics. The code includes library calls for `BIEN` and `tidyverse`, a query to count records in `view_full_occurrence_individual`, and a query to break down occurrence types.
- Console:** Displays the R version (4.4.2), copyright information, and a message about natural language support.
- Environment:** Shows the `total` variable with 1 observation.
- Files:** Lists files in the project directory, including `.gitignore`, `.Rhistory`, `1_BIEN_stats.R`, `2_BIEN_metadata.R`, `3_UNESCO_use_case.R`, `4_BIEN_complete_citations.R`, `BIEN_4.2_update_ms.Rproj`, `citations`, `figures`, `LICENSE`, and `output`.

A red arrow points to the top toolbar, highlighting the absence of a 'Git' tab, which is the subject of the text 'Should have a “Git” tab in RStudio'.

No Git tab?

Tools → Global Options → Git/SVN

Is the git executable found?

Windows: C:/Program Files/Git/bin/git.exe

Restart RStudio after changing it

Other common problems

"Git not found" → restart RStudio, then restart the computer.

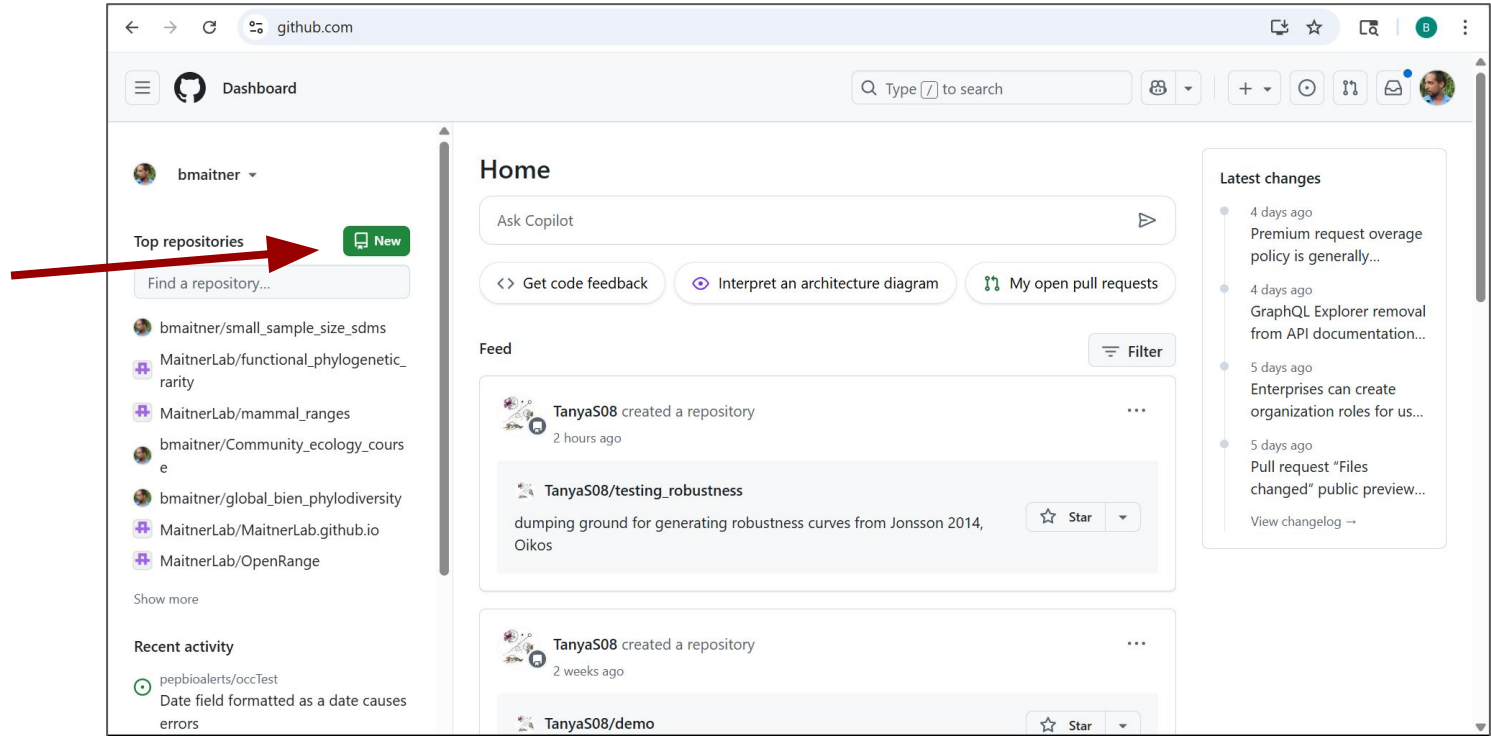
Windows, packages won't build → Rtools

Mac, packages won't build → `xcode-select --install`

Still stuck? Flag me down

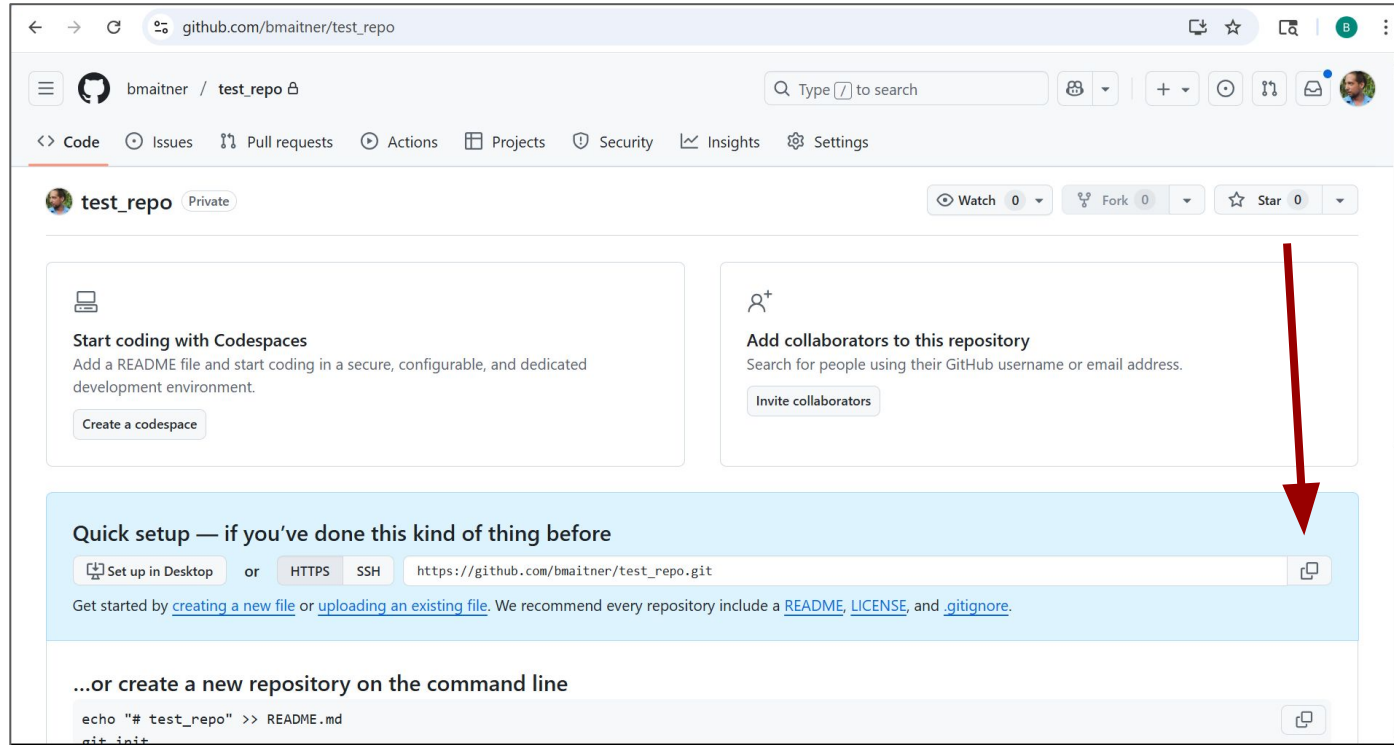
Make sure it works

Go to your Github page and make a new repo



Make sure it works

Copy the URL



The screenshot shows the GitHub interface for a repository named 'test_repo' by user 'bmaitner'. The repository is private. The page includes navigation tabs for Code, Issues, Pull requests, Actions, Projects, Security, Insights, and Settings. Below the repository name, there are buttons for Watch (0), Fork (0), and Star (0). The main content area has two cards: 'Start coding with Codespaces' and 'Add collaborators to this repository'. Below these is a blue 'Quick setup' section with the text 'Quick setup — if you've done this kind of thing before'. It provides a link to 'Set up in Desktop' or 'HTTPS' or 'SSH' with the URL 'https://github.com/bmaitner/test_repo.git'. A red arrow points to the copy icon next to the URL. Below the URL, it says 'Get started by [creating a new file](#) or [uploading an existing file](#). We recommend every repository include a [README](#), [LICENSE](#), and [.gitignore](#).' At the bottom, there is a section for '...or create a new repository on the command line' with a code block showing 'echo "# test_repo" >> README.md' and 'git init'.

github.com/bmaitner/test_repo

bmaitner / test_repo

Type to search

<> Code Issues Pull requests Actions Projects Security Insights Settings

test_repo Private

Watch 0 Fork 0 Star 0

Start coding with Codespaces
Add a README file and start coding in a secure, configurable, and dedicated development environment.
[Create a codespace](#)

Add collaborators to this repository
Search for people using their GitHub username or email address.
[Invite collaborators](#)

Quick setup — if you've done this kind of thing before
[Set up in Desktop](#) or [HTTPS](#) or [SSH](#) `https://github.com/bmaitner/test_repo.git`
Get started by [creating a new file](#) or [uploading an existing file](#). We recommend every repository include a [README](#), [LICENSE](#), and [.gitignore](#).

...or create a new repository on the command line

```
echo "# test_repo" >> README.md  
git init
```

Make sure it works

In RStudio:

- File > New Project...
- Select “Version Control”
- Select “Git”
- Paste the copied URL into the “Repository URL” box
- Pick a directory for it to be created in
- Click “Create Project”

Cloning worked. Pushing won't

Downloading a public repo: no login

Sending changes back: requires login

GitHub stopped taking account passwords in 2021

- Requires authentication

Get a token

```
usethis::create_github_token()
```

Opens GitHub with the right boxes ticked.

Name it, generate, copy it now (you won't see it again)

Store your token

```
gitcreds::gitcreds_set()
```

Paste when prompted. Once per machine.

```
usethis::git_sitrep()
```

Checks everything at once.

Treat the token like a password. Never use in a script.

Multiple repositories

Your Github repo: where your assignments will live.

Can keep public, or make private and grant me access

The course repo: lectures, scripts, data

github.com/bmaitner/biometry_course

Can browse online or pull everything using Git

Note: For grad students, one repo per paper/thesis chapter usually works well

Next time:

Before class: read 1.4 - 1.6

During class:

- Discuss 1.4-1.6
- Work through 1.7